

Hilbert Trace Classification for Second-Order Besov Maximal Regularity

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Status. Candidate major partial result, likely valid, pending human review. The packet gives a full Hilbert classification when $D(A) \hookrightarrow D(B)$, an explicit strongly commuting spectral formula, and the complete self-adjoint fractional-damping phase diagram. The source's unrestricted Banach/noncommuting problem remains open.

1 Source question

Batty, Chill, and Srivastava study Besov maximal regularity for

$$u'' + Bu' + Au = f, \quad u(0) = u_0, \quad u'(0) = u_1,$$

on a Banach space X [1]. For $1 \leq p < \infty$, $1 \leq q \leq \infty$, and $0 < \theta < 1/p$, their trace space is

$$(X, D(B), D(A))_{B_{pq}^\theta} := \{(u(0), u'(0)) : u \in W^{2,p}(0, T; X), \\ u'', Bu', Au \in B_{pq}^\theta(0, T; X)\}.$$

At the end of Section 5, Example 5.3, printed page 13, they write:

“The identification of the trace space $(X, D(B), D(A))_{B_{pq}^\theta}$, even for particular choices of X , B and A , is left as an open problem.”

Maximal regularity in interpolation spaces

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We point out that in the particular case $p = 1$ there is no restriction on the value of $\theta \in (0, 1)$.

The identification of the trace space $(X, D_B, D_A)_{B_{pq}^\theta}$, even for particular choices of X , B and A , is left as an open problem.

Figure 1: The open problem in the original source PDF, printed page 13.

The earlier version of this packet identified the critical self-adjoint case $B = \alpha A^{1/2}$ for $p = q = 2$. The present upgrade classifies a much larger Hilbert class and, in particular, every fractional damping $B = \alpha A^\varepsilon$ with $0 \leq \varepsilon \leq 1$.

2 Definitions

Let H be a complex Hilbert space, $T > 0$, and $0 < \theta < 1/2$. Let A and B be densely defined closed operators on H , and assume throughout the general theorem that

$$\mathbf{D}(A) \hookrightarrow \mathbf{D}(B) \quad \text{continuously for their graph norms.} \quad (1)$$

Define

$$\mathcal{E}_\theta(0, T; A, B) := \{u \in W^{2,2}(0, T; H) : u'', Bu', Au \in H^\theta(0, T; H)\}.$$

Here $H^\theta = B_{22}^\theta$, with equivalent norms. The trace range

$$\mathrm{Tr} \mathcal{E}_\theta := \{(u(0), u'(0)) : u \in \mathcal{E}_\theta(0, T; A, B)\}$$

has the quotient norm induced by the graph norm of $\mathcal{E}_\theta$.

For $\tau \in \mathbb{R}$, let Q_τ be the positive self-adjoint operator associated with the closed quadratic form on $\mathbf{D}(A)$

$$q_\tau[x] := (\tau^4 + \tau^2 + 1)\|x\|^2 + \tau^2\|Bx\|^2 + \|Ax\|^2. \quad (2)$$

The embedding (1) makes this a closed form with a fixed form domain. The fixed-domain form-resolvent identity and $\|Bx\| \lesssim \|Ax\| + \|x\|$ show that $\tau \mapsto Q_\tau^{-1}$ is norm-continuous. Since $Q_\tau \geq (1 + \tau^2 + \tau^4)I$, the positive Bochner operator integrals

$$K_j := \int_{\mathbb{R}} \tau^{2j} (1 + \tau^2)^{-\theta} Q_\tau^{-1} d\tau, \quad j = 0, 1, \quad (3)$$

converge in operator norm. Both K_j are bounded, positive, and injective. For such a K , the operator range $\mathrm{Ran} K^{1/2}$ carries the norm $\|x\|_{\mathrm{Ran} K^{1/2}} := \|K^{-1/2}x\|$.

3 Main results

Theorem 1 (Exact Hilbert operator-range classification). *Under (1),*

$$\boxed{\mathrm{Tr} \mathcal{E}_\theta = \mathrm{Ran} K_0^{1/2} \times \mathrm{Ran} K_1^{1/2}.} \quad (4)$$

Moreover,

$$\|(u_0, u_1)\|_{\mathrm{Tr} \mathcal{E}_\theta}^2 \asymp \|K_0^{-1/2}u_0\|^2 + \|K_1^{-1/2}u_1\|^2.$$

Thus (3) is a complete trace-space identification for this Hilbert class, without assuming that A, B commute or are self-adjoint.

The formula becomes elementary spectral multiplier theory when A and B are strongly commuting nonnegative self-adjoint operators. Write $\alpha, \beta \geq 0$ for their joint spectral variables and set

$$a = (1 + \alpha^2)^{1/2}, \quad b = (1 + \beta^2)^{1/2}, \quad S = (a + b^2)^{1/2}, \quad r = \frac{a}{S}. \quad (5)$$

Then $0 < r \leq S$. Define joint spectral multipliers M_0, M_1 by

$$M_0^2 = aS(1 + r^2)^\theta, \quad M_1^2 = S(1 + S^2)^\theta. \quad (6)$$

Theorem 2 (Closed formula for commuting self-adjoint pairs). *Suppose, in addition to (1), that A, B are nonnegative self-adjoint and strongly commute. Then*

$$\boxed{\operatorname{Tr} \mathcal{E}_\theta = \operatorname{D}(M_0) \times \operatorname{D}(M_1),} \quad (7)$$

with equivalence between the quotient norm and the two graph norms.

The source paper's fractional damping family now has a complete exponent diagram.

Corollary 1 (All self-adjoint fractional dampings). *Let G be nonnegative self-adjoint on H , let $c > 0$, and put*

$$A = G, \quad B = cG^\varepsilon, \quad 0 \leq \varepsilon \leq 1.$$

Then

$$\operatorname{Tr} \mathcal{E}_\theta = \operatorname{D}((I + G)^{\gamma_0}) \times \operatorname{D}((I + G)^{\gamma_1}), \quad (8)$$

where

$$(\gamma_0, \gamma_1) = \begin{cases} (\frac{3}{4} + \frac{\theta}{2}, \frac{1}{4} + \frac{\theta}{2}), & 0 \leq \varepsilon \leq \frac{1}{2}, \\ (\frac{1+\varepsilon}{2} + \theta(1-\varepsilon), \varepsilon(\frac{1}{2} + \theta)), & \frac{1}{2} \leq \varepsilon \leq 1. \end{cases} \quad (9)$$

The formulas agree at $\varepsilon = 1/2$. In particular:

$$\begin{aligned} \varepsilon = \frac{1}{2}: & \quad \operatorname{D}((I + G)^{3/4+\theta/2}) \times \operatorname{D}((I + G)^{1/4+\theta/2}), \\ \varepsilon = 1: & \quad \operatorname{D}(I + G) \times \operatorname{D}((I + G)^{1/2+\theta}). \end{aligned}$$

4 Proof intuition

The domain embedding turns the finite-interval graph space into an ordinary intersection of three Hilbert-valued Bessel-potential spaces, so one common time extension is available. Fourier transformation in time produces the positive form pencil Q_τ . The displacement and velocity evaluations are bounded maps from this Fourier energy space to H . Their mixed Gram operator is zero by parity, while the two diagonal Gram operators are exactly K_0 and K_1 . The Hilbert operator-range identity then proves Theorem 1, including surjectivity and the quotient norm.

For a commuting self-adjoint pair, the form pencil is scalar on every joint spectral fiber. Its fourth-order polynomial has a slow scale r and a fast scale S . The displacement trace sees the slow scale; the velocity trace sees the fast scale. A two-scale integral lemma converts that observation into the exact multipliers (6). Substituting $B = cG^\varepsilon$ yields the two phase regimes in (9).

5 Proof of the operator-range theorem

Lemma 1 (Common whole-line extension). *The graph space $\mathcal{E}_\theta(0, T; A, B)$ is, with equivalent norms,*

$$H^{2+\theta}(0, T; H) \cap H^{1+\theta}(0, T; \operatorname{D}(B)) \cap H^\theta(0, T; \operatorname{D}(A)). \quad (10)$$

One scalar time-extension operator maps this intersection boundedly into the corresponding whole-line intersection.

Proof. If $u \in \mathcal{E}_\theta$, then $u'' \in H^\theta$ and $u \in W^{2,2}$ imply $u \in H^{2+\theta}(0, T; H)$. Also $u \in H^\theta(0, T; \operatorname{D}(A))$. By (1), $Bu \in H^\theta$; together with $Bu' \in H^\theta$, this yields $Bu \in H^{1+\theta}$ and hence the middle space in (10). The converse follows immediately by differentiation and the closed graph norms.

A finite reflection matching the function and its first derivative, followed by a fixed cutoff, is bounded on $H^s(0, T; Y)$ for every Hilbert space Y and $0 \leq s < 5/2$ in a fixed compact range. The same scalar formula applies with $Y = H, D(B), D(A)$, proving the last assertion. $\square$

Let $\mathfrak{X}_\theta$ denote the resulting whole-line Hilbert space. With the unitary Fourier transform, its squared norm is equivalent to

$$\|u\|_{\mathfrak{X}_\theta}^2 = \int_{\mathbb{R}} (1 + \tau^2)^\theta q_\tau[\widehat{u}(\tau)] d\tau. \quad (11)$$

Fourier inversion defines the bounded trace map

$$T : \mathfrak{X}_\theta \longrightarrow H \times H, \quad Tu = (u(0), u'(0)).$$

The bounds follow by Cauchy–Schwarz in the form norm and $Q_\tau^{-1} \leq (1 + \tau^2 + \tau^4)^{-1} I$.

Relative to the inner product in (11), the adjoints of the two coordinate maps are, up to the unitary Fourier constant,

$$\begin{aligned} (T_0^* h)(\tau) &= (1 + \tau^2)^{-\theta} Q_\tau^{-1} h, \\ (T_1^* h)(\tau) &= -i\tau(1 + \tau^2)^{-\theta} Q_\tau^{-1} h. \end{aligned}$$

Since $Q_{-\tau} = Q_\tau$, the mixed Gram operators vanish, and

$$TT^* = (2\pi)^{-1} \begin{pmatrix} K_0 & 0 \\ 0 & K_1 \end{pmatrix}. \quad (12)$$

For a bounded operator between Hilbert spaces, $\text{Ran } T = \text{Ran}((TT^*)^{1/2})$, with equality of the natural quotient/operator- range norms; this is the standard Hilbert range identity behind Douglas' range theorem [2]. Equation (12) proves the whole-line version of (4). Lemma 1 and restriction transfer both inclusions to $(0, T)$. Since restriction is bounded and the common extension is a bounded right inverse to restriction, their induced quotient norms are equivalent. This proves Theorem 1.

6 The two-scale spectral calculation

Lemma 2 (Weighted two-scale integrals). *Let $0 < \theta < 1/2$ and $0 < r \leq S$. Then, uniformly in r, S ,*

$$\int_{\mathbb{R}} \frac{(1 + \tau^2)^{-\theta}}{(\tau^2 + r^2)(\tau^2 + S^2)} d\tau \asymp \frac{1}{rS^2(1 + r^2)^\theta}, \quad (13)$$

$$\int_{\mathbb{R}} \frac{\tau^2(1 + \tau^2)^{-\theta}}{(\tau^2 + r^2)(\tau^2 + S^2)} d\tau \asymp \frac{1}{S(1 + S^2)^\theta}. \quad (14)$$

Proof. Put $\delta = r/S \leq 1$. After $\tau = rs$, the first integral equals

$$\frac{1}{rS^2} \int_{\mathbb{R}} \frac{(1 + r^2 s^2)^{-\theta}}{(1 + s^2)(1 + \delta^2 s^2)} ds.$$

Dropping the last denominator gives an upper bound. If $r \leq 1$, the remaining integral is uniformly bounded. If $r \geq 1$, split at $|s| = 1/r$: the inner piece has length $O(r^{-1}) = O(r^{-2\theta})$, and on the outer piece use $(1 + r^2 s^2)^{-\theta} \leq r^{-2\theta} |s|^{-2\theta}$. The latter majorant is integrable because $\theta < 1/2$. A lower bound of order $(1 + r^2)^{-\theta}$ follows by restricting to $1/2 \leq |s| \leq 1$. This proves (13).

After $\tau = Ss$, the second integral is

$$\frac{1}{S} \int_{\mathbb{R}} \frac{s^2(1 + S^2 s^2)^{-\theta}}{(s^2 + \delta^2)(s^2 + 1)} ds.$$

For the upper bound use $s^2/(s^2 + \delta^2) \leq 1$ and repeat the preceding split with S . Restriction to $1 \leq |s| \leq 2$ gives the matching lower bound. This proves (14). $\square$

On a joint spectral fiber, (2) is

$$\tau^4 + b^2 \tau^2 + a^2.$$

With S, r from (5),

$$\tau^4 + b^2 \tau^2 + a^2 \asymp (\tau^2 + r^2)(\tau^2 + S^2). \quad (15)$$

Indeed, $a\tau^2 \leq (\tau^4 + a^2)/2$, so adding $a\tau^2$ changes the polynomial by only a universal factor; moreover $S^2 + r^2$ is between S^2 and $2S^2$ and $r^2 S^2 = a^2$.

Lemma 2 says that the two scalar values of K_0, K_1 are comparable to the reciprocals of M_0^2, M_1^2 in (6). Joint spectral calculus therefore turns Theorem 1 into (7), proving Theorem 2.

For Corollary 1, let λ be the spectral variable of G . Up to constants depending only on c , $a \asymp 1 + \lambda$ and $b \asymp (1 + \lambda)^\varepsilon$. If $\varepsilon \leq 1/2$, then $S \asymp a^{1/2}$ and $r \asymp a^{1/2}$. If $\varepsilon \geq 1/2$, then $S \asymp a^\varepsilon$ and $r \asymp a^{1-\varepsilon}$. Substitution in (6), followed by taking square roots, gives exactly (9). This proves the corollary.

7 Why the unrestricted problem is not claimed solved

The embedding (1) is used only at the finite-interval extension step, but it is substantive. Without it, cutting off a reflected extension differentiates the cutoff and produces Bu , whereas the source space assumes only Bu' . A diagonal example shows that a decaying whole-line formula cannot simply be imposed.

Take $H = \ell_2$, let $Ae_n = ne_n$, and let $Be_n = n^3 e_n$. On a fixed finite interval, the constant path $u(t) = u_0$ shows that every $u_0 \in D(A)$ is an admissible displacement trace with zero velocity, independently of B . By contrast, the decaying whole-line two-scale weight would impose a displacement weight of order n^4 in the squared norm, i.e. a strictly smaller domain. Thus the unrestricted finite-interval problem retains joint-domain geometry that the present whole-line pencil does not encode. This packet does not claim that no other general description is possible; it explains why the proved formula is scoped by (1).

8 Verification, novelty, and review scope

Proof audit. The proof separately verifies the interval intersection, the closed form pencil, convergence of both operator integrals, the parity-diagonal Gram operator, the Hilbert range identity, uniform two-scale estimates, and every fractional exponent. The script `code/check_integrals.py` tests the critical and two-scale normalizations over weak, critical, and overdamped regimes; the analytic estimates above are the proof.

Novelty check. Bounded searches covered the run indexes and source corpus, official arXiv, and web search. Queries used arXiv:1404.2967, the exact open sentence, the title/authors, “second order Cauchy trace space”, “operator range”, “quadratic pencil”, “strongly commuting”, “strong damping”, “intermediate damping”, and the fractional exponent pairs. The search found the source

and adjacent maximal-regularity literature, but no primary source stating Theorem 1, Theorem 2, or the phase diagram (9). This is bounded evidence, not a novelty certificate; specialist review remains necessary.

Result type. This is a full classification for the Hilbert class $D(A) \hookrightarrow D(B)$ and for the entire nonnegative self-adjoint fractional-damping family $0 \leq \varepsilon \leq 1$. It remains a major partial answer to the source’s arbitrary Banach, arbitrary p, q , and unrelated-domain problem, so the durable folder remains `solutions/partial`.

Human review recommendation. Check the common extension in Lemma 1, the form-pencil adjoint calculation in (12), the exact operator-range norm, and the two regime transition at $\varepsilon = 1/2$. If those checks pass, preserve the operator-range theorem and phase diagram as the primary results; the earlier critical-damping formula is then a corollary.

References

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